Vision-Sentry AI: Autonomous Industrial Safety & Risk Intelligence

Reducing Workplace Accidents and Infrastructure Costs with Edge-Optimized Computer Vision.

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The Hidden Costs of Manual Monitoring

Human Error: Security guards and floor managers cannot monitor every camera 24/7.

Reactive vs. Proactive: Traditional CCTV only helps *after* an accident has happened.

High Infrastructure Costs: Most AI solutions require expensive cloud servers or $2,000+ GPUs.

Privacy Concerns: Constant recording often violates worker privacy regulation

A 24/7 Digital Safety Officer

- **Spatial Fencing:** Automatically detects when personnel enter "Danger Zones" near heavy machinery.
- **Hardware Optimized:** Runs on your **existing Intel-based PCs**—no expensive upgrades needed.
- **Human-in-the-Loop:** Uses Agentic AI to verify threats before triggering alarms, reducing "Alert Fatigue."
- **Privacy-First:** Anonymizes data to track safety trends without tracking individual workers.

Optimized Performance for Industrial Reliability

Intel OpenVINO Integration: We use **INT8 Quantization** to make AI math 3x faster on standard CPUs.

Sub-Millisecond Inference: Detection happens in real-time, ensuring zero-lag alerts.

Robust Logic: Polygon-based coordinate tracking ensures 99% accuracy in defining restricted areas.

Offline Capability: The system works without an internet connection, keeping your data inside your walls.

Turning Video into Actionable Safety Audits

Automated Risk Heatmaps: See exactly which hours and days are high-risk for your facility.

Staffing Optimization: Use data to decide where and when you need more safety supervisors.

Compliance Documentation: Automatically generate logs for insurance and safety audits.

Scalable Architecture: Easily deployable across multiple factory wings or warehouse sections.

Call to Action

The Goal: Zero accidents through intelligent, low-cost automation.

Phase 1 Pilot: A 1-week setup to map your "Danger Zones" and calibrate the AI.

"Let's turn your passive cameras into active safety assets."